MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics
Class: Class X (CBSE)

WEEKLY CHAPTER TEST – 01
Chapter 13: Surface Areas and Volumes

Time: 45 Minutes
MM: 20

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 objective questions (1 mark each).
2. Section B has 2 Short Answer questions (2 marks each). Section C has 2 questions (3 marks each).
3. Section D has 1 Long Answer HOTS question (6 marks). Use $\pi = \frac{22}{7}$ unless stated otherwise.

SECTION A: MULTIPLE CHOICE QUESTIONS (1 Mark Each) [4 Marks]

1. The diameter of a metallic sphere is 6 cm. It is melted and drawn into a wire of diameter 2 mm. The length of the wire is: **[1]**

(a) 12 m (b) 18 m (c) 36 m (d) 66 m

2. A metallic solid sphere of radius 10.5 cm is melted and recast into small cones, each of radius 3.5 cm and height 3 cm. The number of such cones formed is: **[1]**

(a) 63 (b) 126 (c) 21 (d) 130

3. A solid is hemispherical at the bottom and conical above. If the surface areas of the two parts are equal, then the ratio of its radius and the height of its conical part is: **[1]**

(a) $1 : \sqrt{3}$ (b) $1 : 1$ (c) $\sqrt{3} : 1$ (d) $1 : 2$

4. **Assertion (A):** Three cubes each of volume 8 cm^3 are joined end-to-end to form a cuboid. The total surface area of the resulting cuboid is 56 cm^2.
Reason (R): If n cubes each of volume a^3 are joined end-to-end, the total surface area of the cuboid is $2(2n + 1)a^2$. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

5. A vessel is in the form of a hollow hemisphere mounted by a hollow cylinder. The diameter of the hemisphere is 14 cm and the total height of the vessel is 13 cm. Find the inner surface area of the vessel. **[2]**

6. From a solid cylinder of height 2.4 cm and diameter 1.4 cm, a conical cavity of the same height and same diameter is hollowed out. Find the total surface area of the remaining solid. **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

7. A toy is in the shape of a right circular cylinder with a hemisphere on one end and a cone on the other. The radius and height of the cylindrical part are 5 cm and 13 cm respectively. The radii of the hemispherical and conical parts are the same as that of the cylindrical part. Find the surface area of the toy if the total height of the toy is 30 cm. **[3]**

8. A cylindrical vessel with internal diameter 10 cm and height 10.5 cm is full of water. A solid cone of base diameter 7 cm and height 6 cm is completely immersed in water. Find the volume of water:

(i) displaced out of the cylinder.
(ii) left inside the cylindrical vessel.

[3]

SECTION D: LONG ANSWER / HOTS QUESTION (6 Marks) [6 Marks]

9. A right-angled triangle whose perpendicular sides are 15 cm and 20 cm is made to revolve about its hypotenuse.

(i) Determine the common base radius of the double cone so formed.
(ii) Calculate the total volume of the double cone. (Use $\pi = 3.14$)
(iii) Find the total surface area of the double cone exposed to the exterior.

[6]

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